



Universidade do Minho

Licenciatura em Engenharia Informática

Mestrado Integrado em Engenharia Informática

Inteligência Artificial

Practical Sheet n.º 3

Topic: Formulating Research Problems and Problem Solving using Uninformed Research

There are two jugs with capacities of 4 and 3 liters, without markings, initially empty. The goal is to determine what operations to perform so that one of the buckets contains exactly 2 liters from a tank and the other is empty.

The possible operations are:

- Fill one of the two jugs with water from the tank;
 - Empty one of the jugs into the tank;
 - Transfer the contents from one jug to the other until the first one is completely empty or the other one reaches its maximum capacity.
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- 1) Considering the statement, identify the type of problem involved.
 - 2) Formulate the problem as a research problem indicating the representation of the state, initial (current), state/test objective, operators (name, preconditions, effects), and cost of the solution.
 - 3) Solve the problem by hand using a tree search.
 - 4) Solve this problem using the “depth first” strategy.
 - 5) Solve this problem using the “breadth first” strategy.